

Random experiment.
Elementary event.
The total space of elementary events.

1. Random experiment.

Incidence, fortuitousness or otherwise known as randomness, is inherent in every experiment, as it is always random to a greater or lesser extent.

Probability theory is a branch of mathematics that studies regularities in random experiments, also known as random phenomena.

A random experiment is an experiment that **can be repeated many times under the same conditions** and whose results **cannot be unambiguously predicted**.

Activity

Examples of random experiments:

Particular occurrence of experiment:

2. Random events. T

The primary concept is the term "**elementary event**." (equivalent words are elementary occurrence, case)

For each experiment, **it is necessary to define** what is meant by this concept and then write it down formally, providing examples/occurrences (to avoid doubts about these findings) and providing possible elementary events.

3. The total space of elementary events of a given random experiment.

The set of all elementary events of a given random experiment is denoted by the letter Ω .

In the literature, this is called the **(total) space of random events** for a given random experiment.

For countable sets of random events, we use the **cardinality** of the set of all elementary events, denoted $|\Omega|$.

4. Certain amount of events of a given random experiment.

Within a set of elementary events, we often want to distinguish a specific collection - certain number of events based on one or more specific feature/attribute/value of event.

Random events (short: events) are therefore subsets composed of a certain number of elementary events.

A given random event occurs if and only if there is one or more of the elementary events occurs.

Activity

Examples of random events as the subsets of total space of events:

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Particular occurrence of event: